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Advancements on Helix Theory and the Stra.Tech.Man approach: Towards a new synthesis

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Abstract: This article aims to investigate whether the helix theory and the Stra.Tech.Man approach (strategy-technology-management synthesis) have prospects of analytical cross-fertilisation. After presenting the evolution of helix theory of innovation in three different stages and identifying some of its essential points and received criticisms, it analyses the constituents and theoretical implications of the Stra.Tech.Man approach. It finds that some points of the Stra.Tech.Man analysis, such as the "physiology" of the firm, the "competitiveness web" approach that places the dynamics of business innovation centrally, and the proposal of a micro –meso level policy of Local Development and Innovation Institutions that diagnoses Stra.Tech.Man physiology, constitute enrichment and cross-fertilisation elements of the two theories. Overall, both theories attempt to provide a comprehensive theory of integrated socio-economic development, and their synthesis seems to offer new theoretical implications.

Keywords: Helix theory, Stra.Tech.Man approach, business physiology, innovation creation, socio-economic development, Local Development and Innovation Institution, Competitiveness Web

1. Introduction

The triple helix model provides a substantial account of the development process of the development process of socio-economic systems. Triple helix theory studies the evolutionary and institutional links between academia, government policy, and industry from both a national and a local-regional perspective. A critical feature of the study of helix theory is knowledge creation, absorption, and innovation. According to Carayannis and Alexander (1999), in one of the earliest relevant approaches (although not using the term "helix theory" explicitly), governments, universities, and firms develop "co-opetitive" (Brandenburger and Nalebuff, 1996) strategic partnerships by creating complex dynamic knowledge networks (Åsvoll and Widding, 2011).

More generally, the issue of innovation in contemporary educational institutions seems to gain prominence among scholars and politicians as it seems that the traditional function of the university to produce research is no longer sufficient (Sharif, 2019). It now seems that universities need to be actively involved in value creation by interacting dynamically with business ecosystems and local, regional, and national governments (Zhou, 2008).

At the same time, one approach of innovation creation and adaptation to the dynamics of globalisation for the different socio-economic organisms is the "Stra.Tech.Man" approach. This approach assumes that the synthesis between the strategic, technological, and management spheres and dynamics, of all types of socioeconomic actors, determines the level of development for both the different actors and the development trajectory of the entire socio-economic system (Vlados, 2006). Stra.Tech.Man approach is rooted in the assumption that socio-economic organisms are called upon to answer "ontological" questions concerning these particular spheres:

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- ❖ Strategy corresponds to "where does the current development level of the socio-economic organisation lie, and where does it want to reach, and why?"
- ❖ Technology is "how does a socio-economic organisation synthesise and disseminate its expertise within its environment, and why?"
- ❖ Management is "how does a socio-economic organisation use its limited resources, and why?"

In this context, the Stra.Tech.Man model studies innovation simultaneously at the level of strategy, technology, and management and not just in one of the three "realms" (Amini and Saravi-Moghadam, 2017; Beltramini, 2018). By placing the innovative dynamics of the firm at the centre, Stra.Tech.Man theory also approaches how socio-economic systems shape their "multilevel competitiveness" (Peneder, 2017) and not just at the industry or national level (Huang and Huang, 2018).

Therefore, this article seeks to explore the conceptual and methodological linkages between the helix theory and the Stra.Tech.Man approach. It seeks to determine whether there is a potential for bilateral cross-fertilisation between these two analytical categories and propose an overall theoretical model of development and socio-economic competitiveness enhancement.

2. Methodology and structure

To evaluate the relationships between the two theories, that is, the helix theory and the Stra.Tech.Man approach, a literature review is essential because it can point to the evolution and implications of these two analytical classes. The article follows, more specifically, a methodology that attempts a "semi-systematic" literature review that aims to overview the research area and track development over time while trying to synthesise the evidence (Snyder, 2019). The structure of the article reflects this methodology followed:

- First, it presents the dimensions and advancements of helix theory, finding that there are three stages of development that are, to some extent, overlapping. There is a phase of theoretical foundation, which lasts until approximately 2000, followed by a phase of conceptual expansion, which results in a phase of further "maturation" and enrichment that corresponds to recent contributions. It concludes by discussing some of the essential points in the development and criticism of helix theory. The reader should also notice that the literature uses the terms "helix theory" and "helix model" interchangeably, and this article exploits a similar way of presentation.
- Second, it introduces the theoretical constituents of the Stra.Tech.Man approach to innovation and the ways it attempts to conceive the phenomenon of innovation creation. By doing so, it also suggests ways to embed the Stra.Tech.Man model into the helix theory, and vice versa.
- Third, it concludes with a review of the research by suggesting future directions.

3. Helix theory: Analytical dimensions and evolution

The phase of the theoretical foundation of helix theory begins by the work of Etzkowitz and Leydesdorff (1995), which introduces the triple helix model where the central idea is that universities and industries are now closely interconnected institutional spheres. Governments are taking initiatives to lead academia beyond their traditional roles of education and research and, therefore, to contribute actively to wealth creation. Besides, Etzkowitz (1996) analyses the evolutionary perspective of the triple helix model at the regional level. The evolutionary theory of the firm (Nelson and Winter, 1982) suggests that a regional system of innovation involves heterogeneous firms in an industrial district (Becattini, 1990) and, therefore, triple helix theory

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that theorises co-evolutionary relationships between governmental and academic actors can enrich our conceptualisation of innovation creation.

In this evolutionary interpretation, all socio-economic organisations are undergoing a continuous process of innovational transition ("endless transition"). As Etzkowitz and Leydesdorff (1998, p.206) do suggest, "*there is no fixed end-point to transition, nor are only the countries of the former Soviet Union and Central and Eastern Europe in transition: so are the United States, and Asian, Western European and Latin American countries*." From a critical point of view, Viale and Campodall'Orto (2002) are opposing the usual "fuzziness" in the triple helix metaphor and suggest that the best output of science and technology results from the evolution of a strong interaction of academy-industry relations shaped by market forces and not by intervention of government and public bodies

Etzkowitz and Leydesdorff (2000) offer another fundamental exploration of the triple helix system by distinguishing it between the national systems of innovation and "Mode 2." According to the authors, triple helix differs from the national innovation systems (Lundvall, 1992), in which the dynamics of the firm occupies central role to innovation creation. Concerning Mode 2, which suggests (Gibbons et al., 1994) that a new "mode" of knowledge production emerges, which is interdisciplinary and differs from the traditional "mode" of knowledge (Mode 1), triple helix suggests a model at the level of social structure for the explanation of Mode 2 as an historically emerging structure for the production of scientific knowledge.

From then on, a phase of conceptual expansion seems to begin since the academic community accepts the triple helix model gradually and publishes an increasing number of studies. Baber (2001) analyses the impact of globalisation and the emergence of trans-disciplinary scientific fields on this institutional reconstitution of the triple helix. Shinn (2002), from a critical perspective, suggests that the scientific community concerned with triple helix theory and other approaches should not produce an unproductive "frenzy" but, on the contrary, contribute something enduring to scholarship and practice.

Leydesdorff and Etzkowitz (2003, p.57) try to find whether "the public" can constitute a fourth helix in the relationships between government-university-industry, suggesting that "*the conceptualisation of the public as merely a fourth helix narrows the public into another private sphere, rather than seeing civil society as the foundation of the enterprise of innovation. The ability of individuals and groups to organise freely, to debate and take initiatives without permission from the state, can be considered as a necessary condition for the development of a triple helix dynamics of university-industry-government relations that includes both bottom-up and top-down initiatives.*"

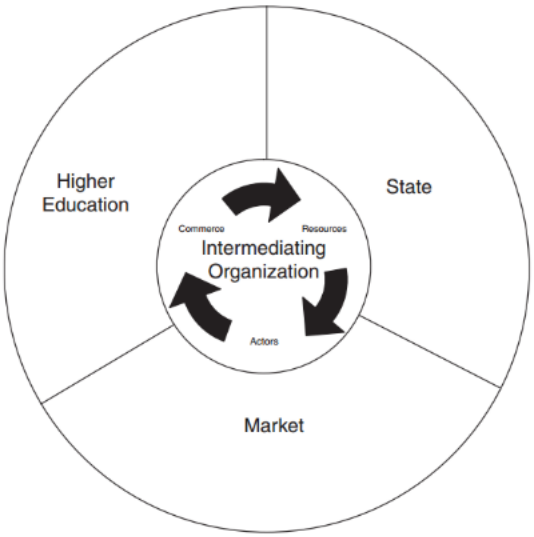
Etzkowitz and Klofsten (2005) propose that the triple helix theory is a model explaining knowledge-based regional development as a set of multi-linear dynamics. The creation of an "entrepreneurial university," whether from an existing academic base or a new foundation, which takes initiatives with government and industry to create a support structure for firm formation and growth, occupies central role for regional development. Etzkowitz and Zhou (2006) argue that the existence of a fourth helix may lead to the loss of triple helix triadic dynamic and, in this sense, propose a "sustainability triple helix of university-public-government" as a complement to the "innovation triple helix of university-industry-government," thereby introducing a missing element into the model while retaining the dynamic properties of a "tertius gaudens."

Johnson (2008) presents how intermediaries support triple helix collaborative research and development by taking a case study of a non-profit organisation in Canada that focuses on pre-commercial development of leading-edge technologies. In a similar methodological concern,

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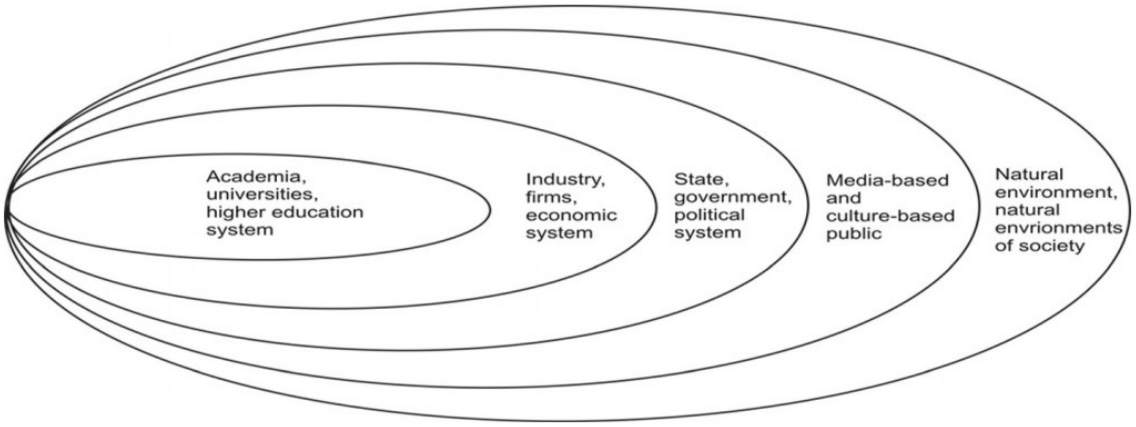
Yuwawutto et al. (2010) study an intermediary organisation in Thailand to strengthen SME innovation by linking university knowledge to the industry while Metcalfe (2010) suggest that intermediary organisations, which are often non-profit organisations, influence the formation of the triple helix relations through the exchange of actors, resources, and commerce (Figure 1).

Figure 1: Model of an intermediating organisation, reproduced by Metcalfe (2010)



Carayannis and Campbell (2009) introduce the existence of a quadruple helix that incorporates the "media-based" and "culture-based" public, which creates an "*innovation ecosystem, well-configured for the knowledge economy and society*." Subsequently, Carayannis and Campbell (2010) introduce the existence of a quintuple helix that incorporates the dimension of "natural environment," as an approach that goes hand-in-hand with sustainable development and social ecology. As they explain, we have to understand "eco-innovation" and "eco-entrepreneurship" on a broader interdisciplinary level of knowledge and innovation (Figure 2).

Figure 2: "The subsystems of the Quintuple Helix model," reproduced by Carayannis et al. (2012)



With the introduction of more helices, the phase of conceptual expansion seems to consolidate even further while more scholars and policymakers start to deal with "triple helix issues." Leydesdorff and Deakin (2011) suggest an "evolutionary model of the triple helix" and explore the level of knowledge of an urban economy according to the support for innovation provided by the civil society. In a similar vein, Etzkowitz (2011) emphasises the role of

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academia in the transition to a knowledge-based society based on a culture of entrepreneurial initiative. Nakwa et al. (2012) stress the role of intermediate organisations in stimulating triple helix networks of SMEs while Fogelberg and Thorpenberg (2012) analyse the development of innovation policy and development organisations in Sweden called "Arenas," modelled around the triple helix innovation theory.

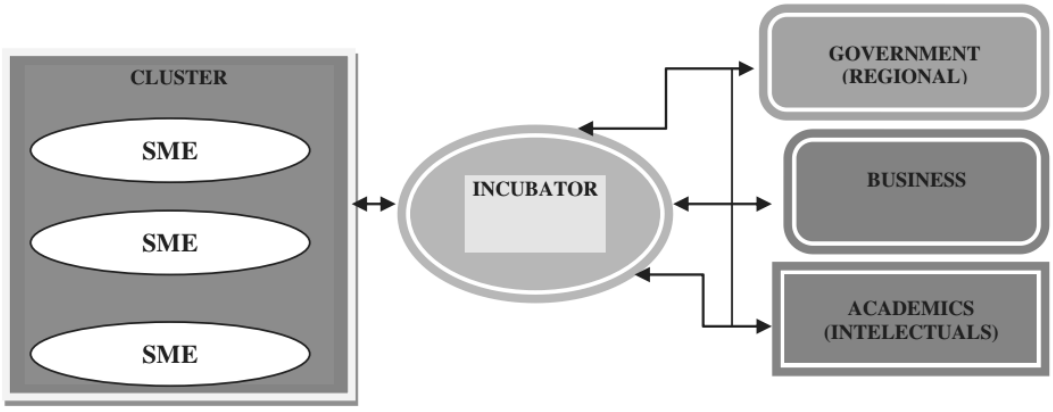
Leydesdorff (2012) introduces the idea of N-tuple of helices by proposing the "algorithmic extension" of the triple helix while explains why misunderstandings in the graphical representations between him and Etzkowitz led them to stop co-authoring. As he explains, Etzkowitz's model is "*neo-institutional by focusing on inter-institutional networking and exchanges*" while his model concerns "*neo-evolutionary mechanisms of exchanges among functions (wealth creation, knowledge production, and normative control)*."

The research by Yang et al. (2012) provides the different outset of triple helix conceptual models:

i. "triple helix" enhances the role of universities in innovation,
ii. "triple helix twins" introduces "the public" into the model while retaining the dynamic "tertius gaudens,"
iii. "quadruple helix" proposes that public reality is crucial for society to assign top-priorities to innovation and knowledge,
iv. "n-tuple helices" are useful in understanding the complexity of innovation and diversity of actors.

In other recent contributions to the helix theory, Rodrigues and Melo (2013) consider the triple helix model for local development policies and find that deficiencies in local helices drive some localities to a lack of required resources to emulate innovative trajectories. Cai (2014) doubts the applicability of the current triple helix in non-Western contexts and less-developed socio-economic systems by suggesting, more specifically, that China needs to adapt its institutional environment to develop further its triple helix innovation institutions. Herliana (2015) also analyses a regional innovation cluster of SMEs enhanced by the successful collaboration between academia, the industry, and the regional government in Indonesia, supported by innovation-based business incubation activities (Figure 3).

Figure 3: Regional innovation clusters of SME: triple helix concept, based on Herliana (2015)



To this end, Wonglimpiyarat (2016), to support the entrepreneurial development in Thailand, proposes university business incubators as intermediaries between academia and industry, which can provide office space, equipment, mentoring services as well as other administrative support to assist the formation of new ventures. Altaf et al. (2018) explore the impact of specific organisations of research and innovation and find that Pakistan does not have a triple helix

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system of innovation and propose better coordination and communication between firms and universities and better monitoring role for the government.

Finally, other recent studies address the micro-aspect of triple helix development. Sá et al. (2018) considers rural entrepreneurship in a Portuguese peripheral region and suggests regional development as an outcome of bottom-up policies by fostering the entrepreneurial culture. Ryan et al. (2018) explore the capabilities of firms to innovate to measure the influence of triple helix programs in Ireland. Cunningham et al. (2018) argue against the macro perspective of quadruple helix models since they fail to conceive value creation activities at the micro-level.

In conclusion, the helix model has been continuously evolving over the last 25 years.

a) In its theoretical foundations, we come across essential elements of evolutionary analysis as helix theory attempts to identify and propose the historical evolution of the three indispensable institutions in modern socio-economic systems, that is, government policy, universities, and firms. We also find fundamental references to regional development issues, as the triple helix institutions bear necessarily locality features.
b) Subsequent approaches attempt to enrich helix theory by introducing more conceptual constituents and interpretative assumptions. A feature of enrichment is the debate on whether more helices "can fit" to the theoretical model. Finally, part of the academic community proposes the existence of quadruple, quintuple, or even n-tuple helices by trying to include all those elements of the socio-economic system that drive innovation and knowledge.
c) Recent studies into the triple helix attempt to shed light on how specific countries and regions develop their innovation in triple helix terms. One criticism is that triple helix methodology concerns mainly developed countries and not all socio-economic systems. However, relevant analyses start increasingly and more systematically to use the triple helix theory while there are also approaches that criticise the macro perspective of helix theory in an attempt to introduce increasingly micro-level theoretical elements. In conclusion, the triple helix theory and its evolutionary foundations appear to constitute an ongoing dialogue that is proving increasingly central to the articulation of economic policy and the dynamic, historical, and evolutionary interpretation of the issue of knowledge and innovation creation.

4. Stra.Tech.Man approach and Helix Theory: Can they be cross-fertilised concepts?

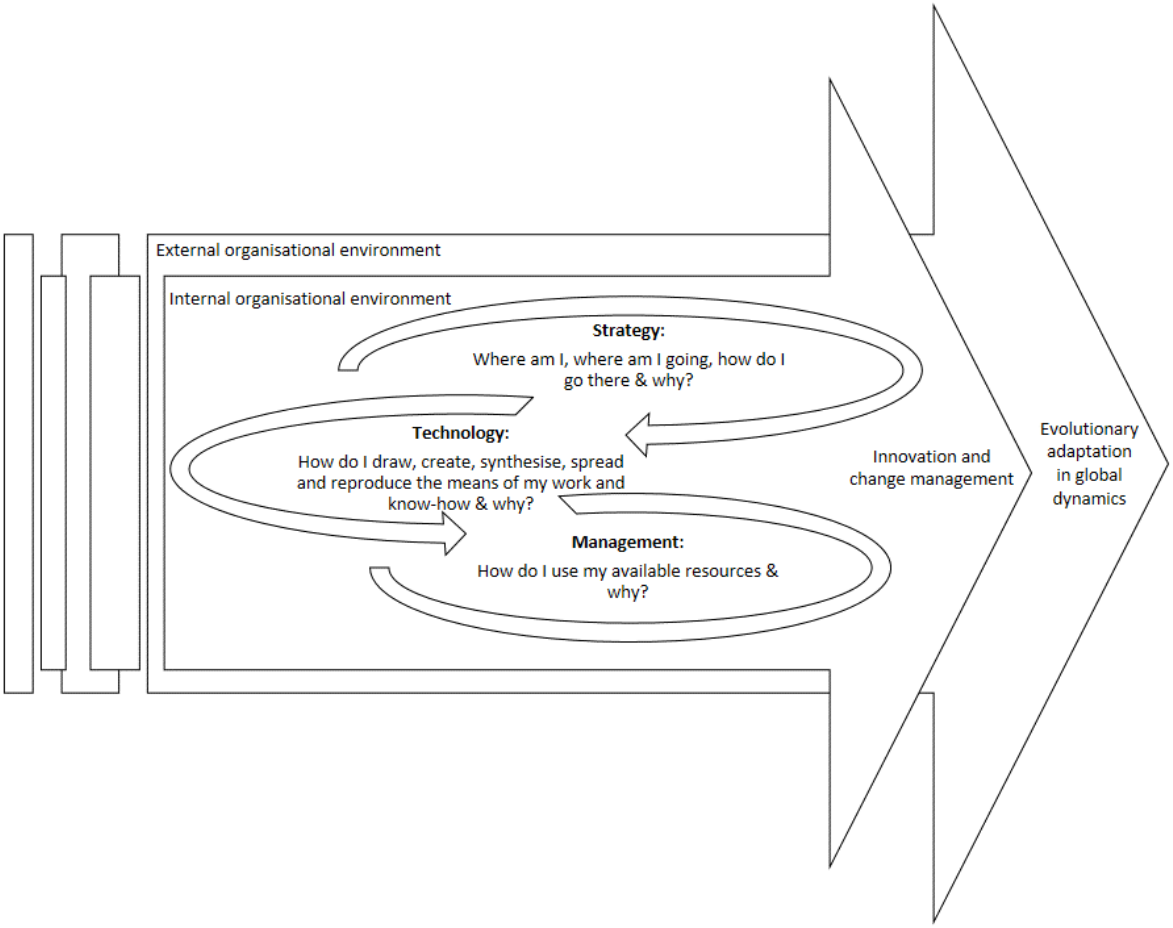
The "micro-elements" of helix theory appear to occupy the thinking of an ever-increasing part of the relevant academic community. To this end, the "Stra.Tech.Man approach" seems to provide a conceptual background that can enrich the helix theory. This section examines the constituents and theoretical implications of the Stra.Tech.Man approach in the light of its synthesis with helix theory.

The Stra.Tech.Man approach considers the action of all "socio-economic organisms" as the result of a synthesis of the dimensions of their strategy, technology, and management (Vlados, 2004). This model suggests that every socio-economic organism responds dialectically, either explicitly or implicitly, in a series of philosophical questions (Figure 4):

- Strategy concerns the "where am I, where do I want to go, how do I go there, and why?"
- Technology corresponds to "how do I compose, utilise, and diffuse the means of my work, and why?"
- Management is about "how do I use my always scarce available resources, and why?"

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Figure 4: The Stra.Tech.Man core of the socio-economic organism.



This dialectical approach defines ultimately the evolutionary trajectory of the socio-economic organism: that is, the specific "physiology" that distinguishes the actor inside an ever-evolving external environment (Vlados and Chatzinikolaou, 2019a). According to the famous passage in Marshall (1890), "*The Mecca of economics lies in economic biology rather than economic mechanics*." In this sense, "Stra.Tech.Man physiology" means that always "environmental limits" exist for the development of a living system. However, since firms are both social and economic organisations, it is common to observe phenomena of "hybridisation" towards "physiologies" that were impossible until recently (Geus, 1997; Penrose, 1995).

Therefore, Stra.Tech.Man physiology determines the evolutionary ability of the socio-economic organism to grow. From a neo-Schumpeterian perspective, this ability corresponds to the innovation of the organisation (Hanusch and Pyka, 2007). Business innovation is primarily an "organic" process (Vlados, 2019a) in the sense that less advanced Stra.Tech.Man physiology results in lower competitiveness and therefore limited range of innovations, rather than a more advanced Stra.Tech.Man physiology. At the same time, the Stra.Tech.Man approach is an evolutionary interpretation of change management. Change management is the unique way that each socio-economic organisation uses its processes and philosophies to develop dialectically to the simultaneous change of its internal and external environment, which is continually being transformed by corresponding internal and external forces of innovation (Vlados et al., 2018).

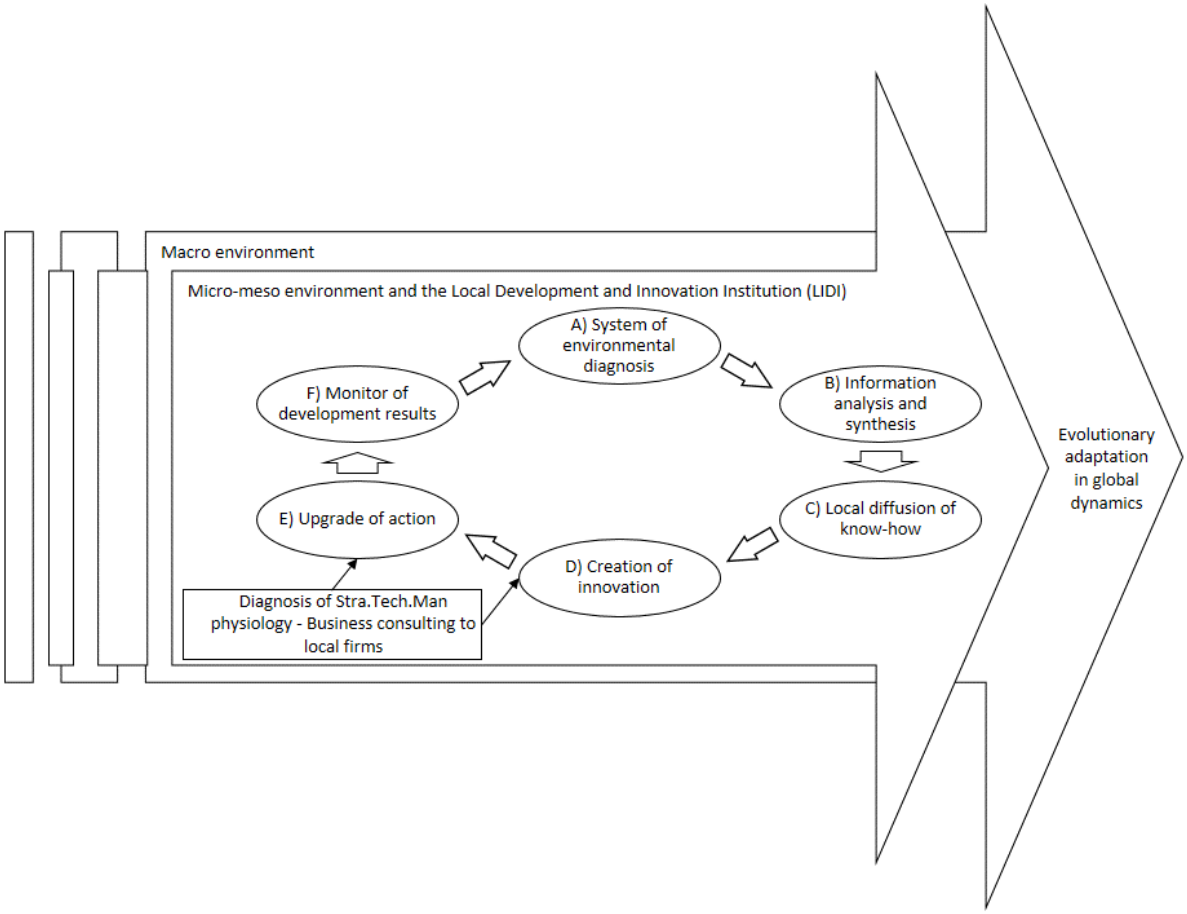
Based on the Stra.Tech.Man approach, a recent study by Vlados and Chatzinikolaou (2019b) used Likert-type questionnaires in a sample of firms in a less-developed business ecosystem. The purpose was through eight strategy questions, eight technology questions, and eight

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management questions to find out how a sample of firms "think" and "act" in terms of Stra.Tech.Man. The respondent had to answer by using personal estimation and evaluation to these Stra.Tech.Man questions, which concerned the "current" and "past five years" experience. The research found and managed to establish that there are no "physiological leaps" for the firms of this ecosystem since the responses were adjacent to each other. For example, a firm could not score maximum technological physiology and minimum strategic physiology: therefore, the general conclusion was that strategy, technology, and management are strictly interdependent spheres in the socio-economic organisation.

In this sense, the research was able to prove that this particular business ecosystem also has "specific physiology" that is primarily due to the innovative and competitive Stra.Tech.Man potential of the firms that is capable of hosting. In this context, it proposed the creation of an intermediate organisation in a less-developed regional business ecosystem in Greece, which aims primarily at fostering local entrepreneurship (Figure 5).

Figure 5: The structure of a Local Development and Innovation Institution (LIDI).



This intermediate organisation called "Local Development and Innovation Institution" (LIDI) has the priority of enhancing micro-competitiveness at the local level. By diagnosing the Stra.Tech.Man physiology of different firms and, therefore, of the overall business ecosystem, it proposes a policy cycle of enhancing micro-competitiveness consisting of six successive steps:

A) System of environmental diagnosis: Initially, it builds an "observatory" by conducting field surveys concerning the level of development of different local-regional firms.

B) Information analysis and synthesis: Subsequently, it analyses this information to connect and coordinate investment opportunities in this less-developed regional business ecosystem.

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C) Local diffusion of expertise: Moreover, it attempts to systematise this expertise and disseminate it to local networks and entities through the creation of relevant forums for local firms.

D and E) Creation of innovation and upgrade of action: It provides business consulting and training services to particular companies on the ground.

F) Monitor of development results: Finally, it articulates a process of collecting data concerning the current level of development of the business ecosystem.

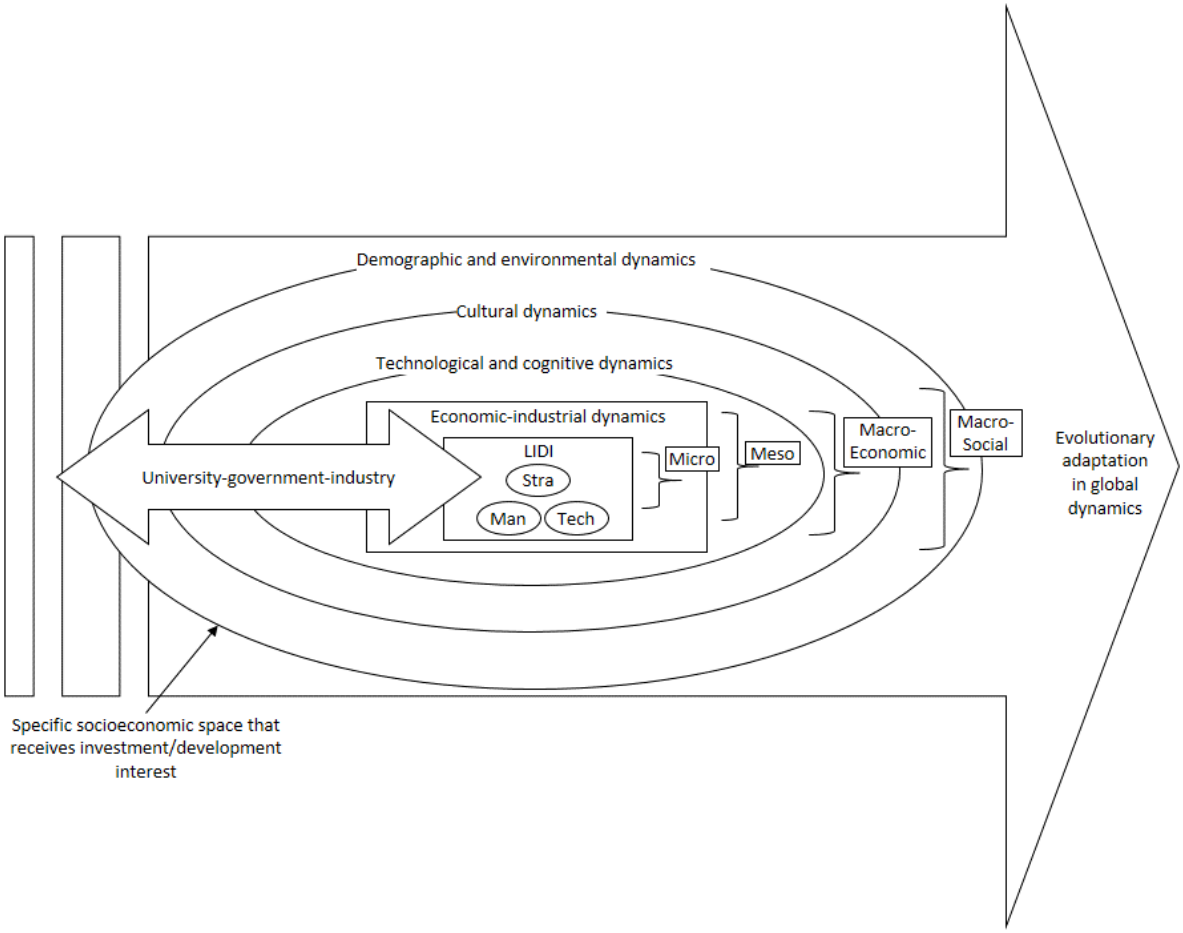
This mechanism can be a local-level institution that interconnects local educational, governmental, and business entities that, until yesterday, at least in the case of Greece and in other less-developed regional business ecosystems, appeared to be uncoordinated. That is, we think that such an institution could be particularly useful if applied in the future in less developed economic ecosystems and economies in crisis, such as Greece (Andrikopoulos and Nastopoulos, 2015; Vlados, 2017).

This policy initiative is primarily of micro-meso origin receiving and delivering flows to the broader macro-environment, suggesting that the evolutionary adaptation of the socio-economic system depends on the level of business development in terms of Stra.Tech.Man. The "LIDI" implies that the innovative capacity of the firms in terms of Stra.Tech.Man determines the overall competitive dynamics of the hosting system, and that innovation is a new application of knowledge that gives better performance today than yesterday (Vlados et al., 2019).

It is a fact today that, in the age of globalisation, which is in a radical restructuring process and where the existing techno-economic paradigm changes form and content (Perez, 2010), the flows between socio-economic systems are multilevel and cross-sectoral. The boundaries of the "industry" are increasingly "perforated" as the productive expressions of different actors in globalisation meet different industries, economic systems, technologies, cultural dynamics, and demographics. According to Vlados (2019b), the socio-economic system in the age of globalisation resembles a "competitiveness web," where the levels of space interact with each other through dynamic, historical, and evolutionary processes (Figure 6).

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Figure 6: The competitiveness web incorporates the triple helix institutions inherently.



By placing the firm's micro-competitiveness at the centre of the system in terms of Stra.Tech.Man, the socio-economic system in the form of "competitiveness web" adapts evolutionarily to global dynamics. In this co-evolving glocal (at the same time, global and local; Cecilia de Burgh-Woodman, 2014) environment, triple-helix dynamics of knowledge and innovation, namely the institutions of university-government-industry, have a decisive influence on the overall prospects of the whole system. Triple helix institutions transform radically both macro-social and macro-economic factors as well as micro-meso factors.

One criticism of the triple helix model is that it lacks micro-foundations and that it does not provide equal treatment for all socio-economic systems. In the competitiveness web approach, which places the firm in Stra.Tech.Man terms at the centre, the triple helix seems to initially enrich its micro-foundations, as it can recognise the micro-behaviours (Gavetti et al., 2012) of all socio-economic actors at the centre of the system. In this sense, the introduction of quadruple and quintuple helices meets competitiveness web in the different macro environments, while the introduction of n-tuple helices, which seek to create a multi-level algorithmic modelling, seems to have no immediate practical application, as the diagnosis of Stra.Tech.Man physiology of local business ecosystems can be the developmental compass for socio-economic systems of competitiveness.

Concerning the non-implementation of the triple helix model in less-developed countries, the micro-competitiveness-centred Stra.Tech.Man approach has usability in different contexts as it can index the transition of the different constituents of the systems ("micro-meso-macro" dynamics). In this sense of the multilevel competitiveness web, the Stra.Tech.Man approach and the triple helix model are cross-fertilised concepts.

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5. Final remarks

This article attempted to study the helix theory and the Stra.Tech.Man approach as parallel concepts in the literature. Although the helix model is older than Stra.Tech.Man, these two theories seem to develop complementarities nowadays. Recognising that knowledge creation and innovation are the most critical developmental features of socio-economic systems in the modern age of the dynamic of globalisation, both theories provide specific methodological tools that enrich our knowledge in this direction.

I. Helix theory, which seems to follow three successive phases of theoretical development, appears to include clear elements of evolutionary analysis as it studies the historical paths and evolution of the three institutions of critical importance, at least in developed socio-economic systems, that is, universities, governmental policies, and industries. At the same time, criticisms on helix theory view the relative inability of the theory to incorporate micro-level constituents more explicitly by also discussing ways to implement the helix model of innovation in the case of less-development countries and business ecosystems (Vlados and Chatzinikolaou, 2019b).

II. To this end, the Stra.Tech.Man approach seems to enrich our understanding of innovation within socio-economic systems systematically. It promotes the idea that innovation in "socio-economic organisms" is the result of a dialectical synthesis between the spheres of strategy, technology, and management by "breaking" the constituents of these fundamental terms into their "simplest" forms. That is, it explores strategy, technology, and management from an "existential" perspective (Sartre, 1960), introducing the idea that the level and potential of development of a socio-economic system (local-regional-national) depends on the "Stra.Tech.Man physiology" of organisations that can host and nurture.

III. The form of the socio-economic system in a "competitiveness web" is yet another enrichment to helix theory. By placing at the multilevel perspective of the socio-economic system the transformations caused by the institutional spheres of the triple helix, it seeks to build an "index" of determining the level of development and competitiveness of the system. Of course, the dynamics of the firm in Stra.Tech.Man terms lie at the heart of the system, while the dynamics of micro-meso level policy interventions to foster local entrepreneurship in terms of diagnosing the particular Stra.Tech.Man physiology is another discussed issue of synthesis between the Stra.Tech.Man and helix theory concepts.

However, this research also has some limitations. Initially, a literature review of helix theory could include a comprehensive bibliometric analysis to find out what contributions disseminate the theory more explicitly (such as in the research made by Lazzeretti et al., 2014, although for the case of the historical development of the clusters concept). This limitation makes sense, as the study attempted to present the focal points of helix theory in order to conclude that micro-foundations are indeed an enrichment element.

Overall, this article suggested the conceptual "amalgamation" between the separate concepts of helix theory and Stra.Tech.Man. Future work could advance the theoretical reflections proposed to the practice level by measuring the impact of the "competitiveness web" elements in the context of a local-regional-national case study. That is, after measuring the Stra.Tech.Man physiology of the business ecosystem to estimate how the technological, cultural, and demographic dynamics affect the system.

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